MILITARY SPECIFICATION

CABLE, SUBMARINE, POWER, CONTROL, AND COMBINATION

1. SCOPE

1.1 Scope.- This specification covers synthetic rubber insulated submarine power and control cable, containing from three to ten conductors, for use in coastal waters for power and control purposes.

1.2 Classification.-

1.2.1 Cables.- The cables shall be the following types, as stipulated (see 6.2).

Type I - Power, three-conductor, No. 6 AWG, for 5000 volts, with ozone resisting compound

Type II - Control, four-conductor, No. 10 AWG, for 600 volts

Type III - Control, ten-conductor, No. 14 AWG, for 600 volts

Type IV - Combined power and control, seven-conductor for 600 volts
(a) Power - Three-conductor, No. 6 AWG
(b) Control - Four-conductor, No. 10 AWG

1.2.2 Protective coverings.- The protective coverings shall be of the following classes, as stipulated (see 6.2).

Class 1 - Asphalt-tar-coated jute and whitewashed jute over tar-dipped armor.
Class 2 - One layer synthetic rubber .109-inch thick.
Class 3 - Two layers synthetic rubber .0625-inch and .1250-inch in thickness respectively.
Class 4 - Two layers synthetic rubber .078-inch and .109-inch in thickness respectively.
Class 5 - One layer polyethylene, pigmented and stabilized, .125-inch thick

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2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on the date of invitation for bids, forms a part of this specification to the extent specified herein:

STANDARD

Federal Standard

601 Rubber, Sampling and Testing

(Copies of Specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification. Unless otherwise specified, the issue in effect on date of invitation for bids shall apply.

ASTM Designation B-6 - Standard Specifications for Slab Zinc (Spelter)

ASTM Designation B-8 - Specification for Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft

ASTM Designation B-33 - Standard Specifications for Tinned Soft or Annealed Copper Wire for Electrical Purposes

ASTM Designation A-90 - Standard Methods of Test for Weight of Coating on Zinc-Coated (Galvanized) Iron or Steel Articles

ASTM Designation B-189 - Specifications for Lead-Coated and Lead-Alloy-Coated Copper Wire for Electrical Purposes

ASTM Designation A-239 - Standard Method of Test for Uniformity of Coating by the Preece Test (Copper Sulfate Dip) on Zinc-Coated (Galvanized) Iron or Steel Articles

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

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National Electrical Manufacturer's Association

WC-3-1964 Rubber Insulated Wire and Cable for the Transmission and Distribution of Electrical Energy

(This publication is also listed under S-19-81, published by Insulated Power Cable Engineers Association.)

(Application for copies should be addressed to the National Electrical Manufacturer's Association, 155 East 44th Street, New York, N.Y. - 10017.)

3. REQUIREMENTS

3.1 Material.- The materials used for the cables shall be as specified herein.

3.1.1 Copper conductors.- Unless otherwise specified, each conductor strand shall consist of tinned or lead alloy coated copper wire. If tinned copper conductors are used, each strand shall meet all requirements of the ASTM Standard B-33 with the exception that the wire shall withstand four cycles of the Sodium Polysulfide test instead of the two specified. If lead alloy coated conductors are used, each strand shall consist of lead alloy coated copper wire meeting the requirements of ASTM Standard B-189.

3.1.2 Armor wire.- The wire shall be of uniform quality throughout, drawn from iron or steel of such quality and composition that when drawn, uniformly annealed and galvanized, the finished wire shall have the properties hereinafter specified. The surface of the wire shall be free from scale, and from injurious flaws, seams and splints. The zinc used for the coating shall be any grade conforming to the requirements of the latest issue of ASTM Standard B-6.

3.1.2.1 Size.- The size of the finished wire shall be expressed as the diameter of the coated wire in decimal fractions of an inch and BWG wire gauge number. The permissible variations over or under nominal diameter shall be as follows:

For wire 0.109 inch or over in diameter -- 0.005 inch

For wire under 0.109 inch in diameter ---- 0.003 inch

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3.1.2.2 Tensile strength.- The galvanized wire shall have a tensile strength of not less than 50,000 pounds nor more than 70,000 pounds per square inch of cross section. The finished wire shall have an elongation of not less than 10 percent in 10 inches. The wire shall not split or break when wrapped in a close helix eight complete turns around a cylindrical mandrel, the diameter of which varies with wire size as follows: sizes .148 inch and smaller, two diameters; sizes over .148 inch to .250 inch inclusive, three diameters; sizes over .250 inch, five diameters. In the wrap test, the zinc coating shall adhere firmly to the wire, so that it does not flake and none of the coating can be removed from the wire by rubbing with the bare fingers.

3.1.2.3 Weight.- The weight of the zinc coating shall be determined by the requirements of ASTM Standard A-90, and shall be not less than the weight outlined in Table I.

TABLE I

Outside diameter of wire in inches	Oz. per sq. ft. of stripped surface
.065	0.50
.083	0.60
.109	0.80
.165	0.90
.203 or over	1.00

3.1.2.4 Tar dip.- Armor wires in class 1 protective coverings shall be tar-dipped. Before application to the cable, the armor wires shall be heated in an oven or other suitable equipment sufficient to drive off all moisture. Wires shall either be precoated before winding on bobbins or coated at the closing cone before entering the closing die. In either case the completely armored cable shall be run through a saturating tank to insure coating on the outer surface. The hot compound of tar shall contain nothing deleterious to the cable core and shall be so prepared as to adhere to the wire and form, when set, a complete and fixed preservative coating.

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3.1.3 Rubber filled fabric tapes.- This tape shall consist of a suitable evenly woven cotton fabric thoroughly impregnated and evenly covered on one or both sides with a colored synthetic or natural rubber compound. The coloring material shall be free of any ingredient that will injure the other component parts of the cable. The colors must be fast under any conditions encountered during manufacture, or subsequent prolonged immersion in water. Tape shall be not less than 0.010-inch in thickness.

3.1.4 Jute.- All jute yarn used for fillers or servings shall be of "First Marks" quality or better, clean, of uniform quality throughout, and free of hardstock or foreign substances. It shall be made of all long jute without waste or cuttings, and shall be evenly spun. Jute yarn shall conform to the minimum values for breaking strength when tested in lengths of 26-inches and the following maximum values for the number of twists per foot listed in Table II.

TABLE II

Wt. per lea (14,400 yard)	Average Breaking Strength Pounds (Minimum)	Twists per ft. (Maximum)
50 lb.	40	21
70 lb.	65	18
100 lb.	85	15
16 lb., 3 ply	45	36 per ply

3.1.4.1 Waterproof jute.- Waterproof jute shall be used for cable core fillers and inner serves. It shall consist of jute as specified above which has been impregnated with suitable waterproofing material prior to making into yarn, such as high melting point waxes or synthetic resins or a combination of the two. The jute shall be highly resistant to deterioration and shall have no contaminating or corrosive effects on other component parts of the cable.

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3.1.4.2 Vacuum impregnated jute.- Vacuum impregnated jute shall be used for outer cable serves. It shall consist of jute yarn as specified above which has been thoroughly dried under vacuum and impregnated with a compound consisting of 50 percent paraffin and 50 percent petroleum asphalt. The paraffin and petroleum asphalt shall be commercially free from dirt and other foreign materials.

3.1.5 Conductor insulating compound.- The properties desired for the insulating compound are low dielectric constant and power factor, low water absorption properties, good physical and aging characteristics, and stability of electrical characteristics after prolonged immersion. The insulating compound shall consist of a properly vulcanized compound containing Buna S (GRS) or other synthetic polymer and such other compounding ingredients as may be necessary. Unless otherwise specified in the detail requirements for specific types of cable, it shall contain no natural crude rubber and be homogeneous in character, tough and elastic. The synthetic polymer and other compounding ingredients shall be free from water soluble constituents and other substances which might prove deleterious to the rubber insulation or the conductor. The insulating compound shall meet the average values tabulated in Table III (a) and (b) when tested as specified in 4.1 and 4.3.

3.1.5.1 Insulating compound for Type I cables.- The properties desired for the insulating compound are low water absorption properties, good physical and aging characteristics, ozone resistance and stability of electrical characteristics after prolonged immersion. The insulating compound which consists of a properly vulcanized compound containing butyl and such other compounding ingredients as may be necessary. Unless otherwise specified in the detail requirements for specific types of cable, it shall contain no natural crude rubber and be homogeneous in character, tough and elastic. The synthetic polymer and other compounding ingredients shall be free from water soluble constituents and other substances which might prove deleterious to the rubber insulation of the conductor. The insulating compound shall meet the average values tabulated in Table III (b) and when tested as specified in 4.1 and 4.3.

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3.1.5.2 Insulating compound for Types II, III and IV cables.- The properties desired for the insulating compound are low dielectric constant and power factor, low water absorption properties, good physical and aging characteristics, and stability of electrical characteristics after prolonged immersion. The insulating compound shall consist of a properly vulcanized compound containing Buna S (GRS) or other synthetic polymer and such other compounding ingredients as may be necessary. Unless otherwise specified in the detail requirements for specific types of cable, it shall contain no natural crude rubber and be homogeneous in character, tough and elastic, The synthetic polymer and other compounding ingredients shall be free from water soluble constituents and other substances which might prove deleterious to the rubber insulation or the conductor. The insulating compound shall meet the average values tabulated in Table III (a) when tested as specified in 4.1 and 4.3.

3.1.6 Protective jacket compound.- The properties desired for the protective jacket compound are good aging, stability of physical characteristics after prolonged immersion, and ability to protect the cable from abrasion or corrosive action. The protective jacket compound shall consist of a properly vulcanized compound containing a Neoprene or other synthetic polymer and such other compounding ingredients as may be necessary. It shall be homogeneous in character, tough and elastic. The jacket compound shall meet the average values tabulated in Table III (a) and (b) when tested as specified in section 4.

3.1.6.1 Polyethylene jacket compound.- If specified under paragraph 6.2, the polyethylene jacket compound shall conform to paragraph 4.1.3.6 of IPCEA Publication No. S-19-81.

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TABLE III (a)
For Type II, III and IV Low Voltage and Control Cables

PROPERTY	UNIT	INSULATION COMPOUND 5/64 Inch wall or over 1st cure	INSULATION COMPOUND 5/64 Inch wall or over 2nd cure	JACKET COMPOUND	TESTED
TENSILE STRENGTH					
Unaged specimens	Min. pounds per sq. inch	900	850	2000	
Aged in oxygen bomb 48 hrs. 70°C.	Min. percent of Orig.	75	70	75	
Aged in oven, 96 hrs. 70°C.	Min. percent of Orig.	75	70	75	
ELONGATION					
Unaged specimens	Min. percent	350	325	400	
Aged in oxygen bomb 48 hrs, 70°C.	Min. percent of Orig.	75	70	—	
Aged in air oven 96 hrs. 70°C.	Min. percent of Orig.	70	65	—	
MODULUS					
Unaged specimens, 200 percent elong.	Min. pounds per sq. inch	300	285	—	
Unaged specimens, 300 percent elong.	Min. pounds per sq. inch	—	—	1300	
COMPRESSION CUTTING					
Unaged specimens	Min. pounds	350	285	750	
Aged in bomb 48 hrs. 70°C.	Min. percent of Orig.	80	75	80	
Aged in water 7 days 70°C.	Min. percent of Orig.	80	75	70	
CORROSION TEST					
7 days 70°C. Increase in res.	Max. percent	5	—	—	
MECHANICAL WATER ABSORP.					
7 days 70°C.	Max. milligram per sq. inch	20	—	—	
ELECTRICAL WATER ABSORPTION					
Dielectric constant, 24 hrs. 70°C.	Max.	3.2	—	—	
Dielectric constant, 7 days 70°C.	Max. percent increase	5.0	—	—	
Dielectric constant, 28 days 70°C.	Max. percent increase	10.0	—	—	
Power factor 7 days 70°C.	Max.	0.0125	—	—	
TEAR TEST	Min. pounds per sq. inch			40	

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TABLE III (b)

Physical and Electrical Properties of Butyl Insulation Compound for Type I Cable

PROPERTY	UNIT	INSULATION COMPOUND 5/64 Inch wall or over 1st cure	JACKET COMPOUND	TESTED
TENSILE STRENGTH				
Unaged specimens	Min. pounds per sq. inch	800	2000	
Aged in oxygen bomb 48 hrs. 70°C	Min. percent of Orig.	80	75	
Aged in oven, 96 hrs. 70°C.	Min. percent of Orig.	85	75	
ELONGATION				
Unaged specimens	Min. percent	400	400	
Aged in oxygen bomb 48 hrs. 70°C	Min. percent of Orig.	80	—	
Aged in air oven 96 hrs. 70°C	Min. percent of Orig.	80	—	
MODULUS				
Unaged specimens, 200 percent elong.	Min. pounds per sq. inch	350	—	
Unaged specimens, 300 percent elong.	Min. pounds per sq. inch	—	1300	
COMPRESSION CUTTING				
Unaged specimens	Min. pounds	350	750	
Aged in bomb 48 hrs. 70°C.	Min. percent of Orig.	80	80	
Aged in water 7 days 70°C.	Min. percent of Orig.	80	70	
CORROSION TEST				
7 days 70°C. Increase in res.	Max. percent	2.0	—	
MECHANICAL WATER ABSORP.				
7 days 70°C.	Max. milligram per sq. inch	15.0	—	
ELECTRICAL WATER ABSORPTION				
Dielectric constant, 24 hrs. 70°C	Max.	4.5	—	
Dielectric constant, 7 days 70°C	Max. percent increase	5.0	—	
Dielectric constant, 28 days 70°C	Max. percent increase	10.0	—	
Power facter 7 days 70°C.	Max.	.035	—	
TEAR TEST	Min. pounds per sq. inch		40	

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3.1.7 Tar and asphalt compound.- This compound shall be composed of coal tar blended with natural asphalt. The melting point of this compound shall be not less than 120°F. as determined by the ball and ring test.

3.1.8 Bronze tape.- Bronze tape shall be of a high quality material and shall be 5 mils nominal in thickness and 1-inch in width and not less than 25-percent overlap.

3.1.9 Shielding tape.- Shielding tape shall be of a semi-conducting non-metallic material suitable to the use specified in 3.2.1.2 and to the method of application specified in 3.2.2.

3.1.10 Metal tape shields.- Metal tape shields shall apply to type IV and shall be in accordance with WC-3-1964, Part 4, Paragraph 4.1.1.1.

3.1.11 Marker tape.- Marker tape shall consist of a thin tape, approximately 1/8-inch wide, of cotton or cellulose, laid directly under the outer sheath. There shall appear on one side of the tape, at intervals of approximately 1-foot, the following printed information:

Name of Manufacturer
Name or location of plant
Date of Manufacture (year)
Specification MIL-C-82403
Serial number

3.2 Construction.-

3.2.1 Conductors.- All conductors shall be of soft annealed copper wire, tinned or lead alloy coated. All wire and coatings shall meet the requirements of 3.1.1. When repairs of joints are made in the conductor, the work shall be performed in such a manner that the joint and adjacent conductor is as strong and durable, electrically and mechanically, as the remainder of the conductor. Conductor details shall be as follows:

(a) Type I.- Type I cable shall be power cable, three conductor No. 6 (26,250 CM), for 5000 volt service, each conductor consisting of seven strands of 0.0612-inch diameter tinned or lead alloy coated copper wire.

(b) Type II.- Type II cable shall be control cable, four conductor No. 10 (10,380 CM), for 600 volt service, each conductor consisting of seven strands of 0.0385-inch diameter, tinned or lead alloy coated copper wire.

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(c) Type III.- Type III cable shall be control cable, ten conductor No. 14 (4,107 CM), for 600 volt service, each conductor consisting of seven strands 0.0242-inch diameter tinned or lead alloy coated copper wire.

(d) Type IV.- Type IV cable shall be combined power and control cable, seven conductor, for 600 volt service. The three power conductors shall be No. 6 (26,250 CM), each conductor consisting of seven strands of 0.0612-inch diameter tinned or lead alloy coated copper wire. The four control conductors shall be No. 10 (10,380 CM), each conductor consisting of seven strands of 0.0385-inch diameter tinned or lead alloy coated copper wire. The control conductors shall be put up as shielded, twisted pairs to eliminate magnetic and electrostatic interference between power and control conductors.

3.2.1.1 Stranding.- All conductors shall be concentrically stranded. The number of conductors and the number and diameter of the wires forming each conductor shall be as specified in 3.2.1. Care shall be taken when stranding to avoid abrasion of the wire and its coating, and such abrasion shall be deemed cause for rejection.

3.2.1.2 Shielding.- When specified (see 6.2), prior to application of conductor insulation a semicon non-metallic tape shall be applied over each conductor for the purpose of eliminating voids between the conductor and the inner surface of the insulation and reducing corona cutting on the inner surface of the insulation.

3.2.2 Conductor insulation.- When no conductor shielding tape has been applied the insulating compound shall be applied concentrically about the stranded conductor and shall fit tightly thereto. When conductor shielding tape has been applied, in conformance with 3.2.1.2, the insulating compound shall be applied concentrically about the stranded and taped conductor and the tape shall be vulcanized to the underside of the insulation. While there shall be a pronounced adhesion of the insulation or the insulation and tape to the conductor, this adhesion shall not be sufficient to prevent the insulation or insulation and tape from being stripped from the conductor when splicing or terminating. In any case the dimensions of the insulation shall conform to the following limits for the type cables classified in 1.2:

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Type	Average Wall Thickness 64th (decimal) of Inch	Minimum Wall Thickness Decimal of Inch
I	12 (0.188)	0.170

(The insulation shall show no cracks after 3 hours exposure to an ozone concentration of not less than 0.010 nor more than 0.015 percent by volume.)

II	5 (0.078)	0.070
III	5 (0.078)	0.070
IV - (Power)	6 (0.094)	0.085
IV - (Control)	5 (0.078)	0.070

3.2.3 Conductor color coding.- Each conductor shall be color coded by application of a fabric tape or by printed color coding, as specified. If fabric tape is used, this tape shall conform to 3.1.3 and shall be applied after vulcanization and upon completion of the preliminary electrical tests, with a suitable lay and overlap of not less than twenty percent of the width of the tape. If printed color coding is used, the color name and color number shall be printed on the conductor insulation at suitable intervals. Unless otherwise specified, the colors used shall be in accordance with the following:

Type I - Three conductor power cable-Red, white and black

Type II - Four conductor control cable-White, red, blue and black

Type III- Ten conductor control cable-Red, white, blue, green, orange, black, one red with black tracer, one white with black tracer, one green with black tracer, one orange with black tracer

Type IV - Three conductor power cable-Red, white, and black
and
Four conductor control cable-Orange, green, blue, and white with black tracer

3.2.4 Cabling.- The required number of taped conductors shall be laid up spirally with a suitable direction of lay. The required number of taped conductors shall be in accordance with Part 5, paragraph 5.2 of IPCEA, S-19-81. All voids shall be filled with waterproof jute conforming to 3.1.4 and 3.1.4.2, to form a smooth round core, over which a rubber-filled black tape conforming to 3.1.3 shall be applied with a minimum overlap of 20-percent of the width of the tape. In four conductor cable, the blue and white taped conductors shall be laid up so as to be opposite or non-adjacent.

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3.2.4.1 Protective coverings.- Protective coverings shall be Class 1, 2, 3, 4, or 5, as described in paragraphs 3.2.4.2 through 3.2.4.5. Class 2, 3, and 4 synthetic rubber jackets shall be cured in a mould or continuous lead sheath extruded directly over the jacket and removed after vulcanization. The compound shall be applied concentrically about the cable in such a manner that it will fit tightly, thereto filling the interstices of the armor, and be free of holes, lumps, depressions, or objectionable waviness, and the finished cable shall present a smooth, dense, shiny appearance. All substances cited in paragraphs 3.1.4.2 through 3.2.4.5 shall conform to the applicable material of paragraphs 3.1.1 through 3.1.8.

3.2.4.2 Class 1.- Class 1 protective covering shall consist of the following in sequence:

(a) One layer of 100 pound waterproof jute.

(b) One layer of 16/3-ply waterproof jute.

(c) When specified (see 6.2), a bronze tape shall be applied over the assembled core (with a 25-percent minimum lapover), followed by a jute bedding, galvanized steel armor, one layer of 100 pound vacuum impregnated jute, one layer of 16/3-ply vacuum impregnated jute and applications of hot tar and asphalt compound shall be made over the armor and final jute serve and the completed cable treated with whitewash.

(d) Galvanized steel armor of size specified in procuring documents (see 6.2) and tar-dipped in accordance with 3.1.2.4.

(e) One layer of 100 pound vacuum impregnated jute.

(f) One layer of 16/3-ply vacuum impregnated jute.

(g) Applications of hot tar and asphalt compound shall be made over the armor and final jute serve and the completed cable treated with whitewash.

3.2.4.3 Class 2.- Class 2 protective covering shall consist of the following in sequence (Note: Used with Types II and III cables only):

(a) One layer of 100 pound waterproof jute.

(b) One layer of 16/3-ply waterproof jute.

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(c) When specified (see 6.2), a bronze tape shall be applied over the assembled core (with a 25-percent minimum lapover), followed by a jute bedding, galvanized steel armor as covered below.

(d) Galvanized steel armor of size specified in procuring document (see 6.2).

(e) One layer of a high quality synthetic rubber jacket conforming to 3.1.6, average thickness to be 7/64-inch (.109) and minimum thickness 0.098-inch.

3.2.4.4 Class 3.- Class 3 protective covering shall consist of the following in sequence (Note: Used with Type I and IV cables only):

(a) One layer of 100 pound waterproof jute.

(b) One layer of 16/3-ply waterproof jute.

(c) When specified (see 6.2), a bronze tape shall be applied over the assembled core (with a 25-percent minimum lapover), followed by a jute bedding galvanized steel armor as covered below.

(d) Galvanized steel armor of size specified in procuring documents (see 6.2).

(e) One layer of high quality synthetic rubber inner jacket not less than 0.0625-inch in thickness.

(f) Two layers of 10/2/3 seine twine.

(g) One layer of high quality synthetic rubber outer jacket with average thickness of 0.1250-inch.

3.2.4.5 Class 4.- Class 4 protective covering shall consist of the following in sequence: (Note: Used with Types I and IV cables only):

(a) One layer of 100 pound waterproof jute.

(b) One layer of 16/3-ply waterproof jute.

(c) When specified (see 6.2), a bronze tape shall be applied over the assembled core (with a 25-percent minimum lapover), followed by a jute bedding, galvanized steel armor as covered below.

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(d) Galvanized steel armor of size specified in procuring document (see 6.2).

(e) One layer of high quality synthetic rubber inner jacket not less than .070-inch thick with average thickness of 5/64 (.078) inch.

(f) Two layers 10/2/3 twine.

(g) One layer of high quality synthetic rubber outer jacket not less than .098-inch thick with average thickness of 7/64 (.109) inch.

3.2.4.6 Class 5.- Class 5 shall be the same as class 2 except polyethylene as stipulated in 1.2.2.

3.2.4.7 Protective covering details.-

3.2.4.7.1 Inner jute serves.- Inner jute serves shall be of waterproof jute meeting the requirements of 3.1.4 and 3.1.4.1. Serves shall be in two layers with suitable reverse directions of lay. The ratio of the length of lay to pitch diameter of each serve shall be not less than three or more than four and one-half, and the number of strands employed shall be such that the close coverage is obtained. The servings shall be so applied that they form a smooth compact layer free from lumps or bunching.

3.2.4.7.2 Armor.- The armor shall consist of tar dipped galvanized steel wire, meeting all requirements of 3.1.2, 3.1.2.1, 3.1.2.2, 3.1.2.3, 3.2.2.4, and Table I. The diameter of wire shall be as specified (see 6.2). It shall be applied spirally in a suitable direction of lay. The ratio of the length of lay to pitch diameter shall be not less than eight or more than 10, and the number of armor wires shall be such as to insure not less than 95 percent coverage. The wire shall be so preformed that it will remain in proper position and produce maximum flexibility of the finished cable. All joints in the armor shall be welded and no weld shall be within 12 feet of another weld. All joints shall be thoroughly cleaned after welding and all requirements specified for the original wire.

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3.2.4.7.3 Outer jute serves.- Outer jute serves for class 1 protective coverings shall be of vacuum impregnated jute meeting the requirements of 3.1.4 and 3.1.4.2. The first serving shall be applied with a lay in the reverse direction from the lay of the armor. The second serving of jute shall be applied with a lay in the reverse direction from the lay of the first jute serving. The ratio of the length of lay to the pitch diameter of each serve shall be not less than three or more than four and one-half, and the number of strands employed shall be such that very close coverage is obtained. The serving shall be so applied that it forms a smooth compact layer free from lumps or bunching. An application of hot tar and asphalt compound meeting the requirements of 3.1.7 shall be made over each layer, followed by a thorough coating of whitewash, as specified for class 1 coverings in accordance with 3.2.4.1.

3.2.4.7.4 Synthetic rubber jacket.- The synthetic rubber jacket for class 2, 3 and 4 protective coverings shall be made of a high quality synthetic rubber compound meeting all requirements of 3.1.6. The jacket shall be curved in a mould or continuous lead sheath extruded directly over the jacket, and removed after vulcanization. The compound shall be applied concentrically about the cable in such a manner that it will fit tightly thereto filling the interstices of the armor, and be free of holes, lumps, depressions, or objectionable waviness and the finished cable shall present a smooth, dense, shiny appearance.

3.2.4.7.5 Seine twine.- In class 3 and 4 protective coverings a serving of two layers of 10/2/3 seine twine shall be applied over the inner jacket, for the purpose of binding the compound firmly to the armor. The seine twine shall be applied spirally in reverse direction, with approximately seven strands per inch in each layer.

3.2.4.7.6 Lead sheath.- If a lead sheath is specified (see 6.2), it shall be composed of commercially pure lead. It shall be close fitting, cylindrical, of uniform thickness, with a minimum wall thickness as specified for the particular type of cable being manufactured. Die marks caused by an interruption of the continuous process of sheathing the cable in the lead press shall not weaken the sheath. The sheath shall be free from holes, blisters, splits, laminations or other defects.

3.3 Electrical characteristics.-

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3.3.1 Insulation resistance.- The insulation resistance of all wire and cable, when tested under the conditions specified in 4.2, shall have a resistance when corrected to a temperature of 60° F. of not less than the value obtained from the following formula:

$$R = K \operatorname{Log}_{10} \frac{D}{d}$$

Where R = Megohms per 1000 ft. of cable

K = 10,000

D = Specified average outside diameter of insulation

d = Specified diameter of conductor (For stranded conductors 94 percent of diameter over the strand)

3.3.2 Voltage requirements.- All insulated conductors, when tested as specified in 4.5, shall withstand the voltages specified in Table IV.

TABLE IV

Average Wall Thickness (64ths of Inch)	AC Volts	DC Volts
5 (Types II, III, and IV)	7,800	23,400
6 (Type IV)	9,450	28,200
12 (Type I)	20,800	62,400

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(1) Conductor tests:
Conductors smaller than 0.182 inch diameter, use voltages in accordance with Table IV. Conductors 0.182 inch diameter or larger and wall thickness exceeding 12/64 inch the AC voltage shall be increased 2,000 volts and the DC voltage shall be increased 6,000 volts above those shown in Table IV.

(2) Intermediate (Core) test:
Use 90-percent of the voltage specified in Table IV.

(3) Completed cable, and carload length:
Use 80-percent of the voltage specified in Table IV.

(4) All tests, conductor, core, completed cable, and carload lengths:
For wall thicknesses of 6/64 inch or less, DC or AC voltage may be used for a duration of five minutes each. For wall thicknesses of over 6/64 inch, both AC and DC tests shall be applied; duration of AC shall be five minutes, and DC 15 minutes. AC or DC shall apply to car load lengths.

3.3.3 Capacity.- The capacity of all wire and cable, when tested under the conditions specified in 4.2, shall have a capacity to ground at 1000 cycles or DC voltage of not more than the value obtained from the following formula:

$$C = \frac{0.00735K}{\text{Log}_{10}\frac{D}{d}}$$

K = 4.5 for type I cable, 3.0 for type II, III and IV cables

D = Specified average outside diameter of insulation

d = Specified diameter of conductor (For stranded conductors 94 percent of diameter over the strand)

For lengths greater than 500 feet the electrostatic capacitance shall be measured at direct current by a direct comparison method with a standard capacitor. In making these tests all conductors not under test and all shields and armor must be grounded.

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4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. See 6.4 for details on inspection arrangements.

4.2 Classification of tests.- The methods of sampling, inspection, and tests conducted on the cable fall within the following classifications:

(a) Tests on samples.
(b) Electrical tests on all entire lengths.
(c) Approval tests.

4.3 Tests on samples.- Tests on samples shall be conducted either in accordance with IPCEA - NEMA standards (see 4.3.1 and Table V) or in accordance with 4.3.2 through 4.4.3.2 of this specification. Sampling for IPCEA - NEMA tests shall be as indicated in Table V. Samples for tests of 4.3.2 through 4.4.3.2 of this specification shall consist of one 10-foot length of insulated conductor selected by the inspector from each end of at least four coils or reels in each lot of 100,000 feet of conductor or fraction thereof.

4.3.1 Tests in accordance with IPCEA-NEMA standards.- The following Table V specifies sampling and tests which shall be conducted in accordance with IPCEA Publication No. S-19-81 (NEMA Publication No. WC-3-1964).

4.3.2 Approval tests.- Unless the insulation or jacket compound which the contractor proposes to furnish has previously been approved by the purchaser, approval tests shall be conducted prior to award of any contract. The object of these tests is to determine whether the contractor is prepared to furnish a compound which meets the electrical and physical requirements of Section 3 of this specification. These tests may be repeated on one, two or three more samples selected at random by the Government inspector from material offered on any order, to determine whether such material duplicates the properties of the original sample. Tests shall be made at the factory of the contractor. The manufacturer shall state in bid the number of the compound and description, brand name which he proposes to furnish, and no bids will be considered until this particular compound has been tested and approved.

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TABLE V

Test	Sampling-Paragraph's of: IPCEA Pub. No. S-19-81 NEMA Pub. No. WC-3-1964	Test Paragraph's of IPCEA Pub. No. S-19-81 NEMA Pub. No. WC-3-1964
Conductor	6.3.1	6.3.3. and 6.3.4
Thickness		6.4.2 and 6.4.3
Tensile strength	6.4.4 through 6.4.12.4	6.4.13
Tensile stress (Modulus)	6.4.4 through 6.4.12.4	6.4.13
Elongation	6.4.4 through 6.4.12.4	6.14.13
Tear	6.4.4 through 6.4.12.4	6.14.13
Ozone resistance	6.8.1 and 6.8.3	6.8.2. and 6.8.4

4.3.3 Aging test.- Aging test shall conform to paragraph 6.4.14 of IPCEA publication S-19-81 (NEMA Publication No. WC-3-1964).

4.3.3.1 Oxygen bomb.- Test specimens shall be heated in an atmosphere of oxygen at a pressure of 300 pounds per square inch at a temperature of 70°C., plus or minus 1° for a period of 48 hours. The contents of the bomb shall be restricted to the samples representing the lot to be tested. Weight of specimens shall not be over two grams per cubic inch of bomb space. The test specimens shall be suspended vertically in such a manner that they are not in contact with each other or the sides of the bomb.

4.3.3.2 Gear oven.- Test specimens shall be heated at a temperature of 70°C., plus or minus 1°C., for a period of 96 hours, in an oven having free circulation of air. The test specimens shall be suspended vertically in such a manner that they are not in contact with each other or with the sides of the oven.

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4.3.3 Compression cutting test.-

4.3.3.1 Test specimens.- This test shall be made on original samples of the insulation, samples of which have been aged for 48 hours in the oxygen bomb, and samples of which have been aged for seven days in distilled water at 70°C. All test specimens shall be prepared from suitable lengths of the insulated conductor or jacket from which all tape, wax, dust and other extraneous matter have been removed. Specimens to be aged in the bomb shall first be prepared in accordance with 4.3.2.2 and then aged in accordance with paragraph 4.3.2.1. After removal from the bomb, samples shall be given a rest period of not less than 16 nor more than 48 hours. Specimens aged in water may be prepared from samples used for the determination of mechanical water absorption. Immediately after determining the seven-day weight of the sample, it shall be replaced in water at room temperature and kept there until the compression cutting test is made, which shall be within 24 hours. Surface moisture shall be removed prior to making the compression cutting test in the same manner as prescribed for the mechanical water absorption test.

4.3.3.2 Preparation.- Test specimens of conductor insulation shall be prepared by removing a suitable length of the insulation from the conductor as a tube, and cutting it into lengths of approximately 1.25 inches. Each length shall then be cut in half longitudinally, the two halves forming one test specimen. Before dividing the specimen longitudinally, the outside diameter of the insulation shall be measured at right angles to the proposed longitudinal cut. The thickness of the specimen shall be considered to be one-half of the difference between the outside diameter of the insulation and the diameter of the conductor. Test specimens of insulating or protective jackets shall consist of sections of the jacket approximately 1.25 inches in length, 0.75 inch in width and buffed to a uniform thickness as near 0.100 inch as the dimensions of the sample will permit. All heavy corrugations and inserted reinforcing materials shall be removed during the buffing process.

4.3.3.3 Testing machine.- The testing machine shall consist of a fixed cutting tool and a movable anvil between which the test specimens may be compressed. The cutting tool shall consist of a piece of steel drill rod 1.00 millimeter in diameter, mounted at the apex of an acute angle steel wedge. The anvil shall consist of a steel block 1.000-inch in length. The cutting tool and anvil shall be so mounted and adjusted that their faces will remain parallel throughout the test to within 0.0005-inch. The load shall be applied to the anvil through a calibrated coil spring at a uniform speed (at the end of the spring to which the

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lead is applied) of approximately 1.5 inches per minute. Provision shall be made for recording the load applied and the distance between the anvil and the cutting tool, with a precision of plus or minus 10 pounds and 0.001-inch. The maximum capacity of the spring shall be 1000 pounds unless the load required to shear the specimen exceeds that amount, in which case a spring with a maximum capacity of 2000 pounds shall be employed.

4.3.3.4 Method.- The test shall be made by compressing the specimen between the cutting tool and the anvil, and noting the load in pounds at which rupture occurs. Specimens of the conductor insulation shall be placed with the convex side resting on the anvil in such a position that both ends of the specimen overlap the ends of the anvil, and the cutting tool is parallel with and centered in the concavity formerly occupied by the conductor. The two halves of the specimen shall be compressed successively and the results averaged. Jacket specimens shall be flat on the anvil in such a position that both ends of the specimens overlap the ends of the anvil. Two successive compression cuts shall be made on each jacket specimen approximately 3/8-inch apart and the results averaged. The average load at rupture for each specimen shall be multiplied by the coefficient (K) contained in Table IV which most nearly corresponds with the thickness of the specimen.

TABLE IV

Thickness of Specimen (In.)	(K) For Insulating Compound	(K) For Protective Jacket Compounds
.140	0.90	0.95
.130	.92	.96
.120	.95	.98
.110	.97	.99
.100	1.00	1.00
.090	1.03	1.01
.080	1.07	1.03
.070	1.12	1.06
.060	1.19	1.09
.050	1.27	1.13
.040	1.41	1.20
.030	1.60	
.020	2.07	----

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4.3.4 Copper corrosion test.-

4.3.4.1 Test specimens.- The specimens shall consist of samples of the insulated conductor, approximately six inches in length from which all tape, wax, dust, and other extraneous matters have been removed. Two double test leads of tinned copper wire 0.040-inch in diameter shall be bound firmly with thread to one end of the test specimen, and the lead inserted through four tightly fitting holes in a cork stopper. Two ends of a piece of clean, bare, annealed copper wire, 0.0063-inch in diameter and approximately 30 inches in length shall be soldered to the two test leads and applied spirally around the test specimen as a bifilar winding. The copper wire shall be applied under sufficient tension to insure good contact with the specimen, and in such a manner as to assure separation between adjacent turns. The end of the looped copper wire shall be bound in place with thread. Each specimen shall be placed in an eight inch tube containing a small quantity of distilled water. The quantity of water shall not be sufficient to afford direct contact with the specimen. The stopper shall fit tightly in the test tube to prevent excessive evaporation.

4.3.4.2 Original resistance.- The resistance of the bifilar copper winding shall be measured with a Kelvin bridge with a precision of 0.001 ohm. The measurement shall be made with the tube containing the specimen immersed in a water bath of known temperature (approximately that of the room).

4.3.4.3 Percentage increase.- After original measurement of the resistance, the tube containing the specimen shall be maintained at a temperature of not less than 69°C. or more than 71°C. for a period of seven days. At the end of this period, the resistance shall be measured again at the same temperature as that at the time of the original measurement. The percentage increase in resistance shall not exceed the limit specified for the type of compound under test, and shall be computed from the following formula:

$$\frac{R_2 - R_1}{R_1} X\ 100$$

Where R_1 = Original resistance
R_2 = Corrected resistance after seven days

4.4 Water absorption tests.-

4.4.1 Test specimens.- Test specimens of the conductor insulation for mechanical and electrical water absorption tests shall consist of a suitable length of the insulated conductor. All wax or other substances adhering to the surfaces of test specimens shall be carefully removed, and

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the surfaces thoroughly cleaned by scrubbing with water and a small brush or lintless cloth. Specimens shall be rinsed and wiped dry. Specimens for mechanical water absorption tests shall be suspended in a desiccator or other compartment for a period of not less than 20 nor more than 24 hours. The desiccator or compartment shall be maintained at a temperature of not less than 69°C. nor more than 71°C., and a vacuum maintained therein of not more than three cm. of mercury. Specimens for electrical water absorption tests shall be dried in an oven, exposed to circulating air, at the same temperature, and for the same period. Specimens shall be suspended in such a manner that they do not touch each other or the sides of the desiccator or oven.

4.4.2 Mechanical water absorption test.-

4.4.2.1 Method.- After removal from the desiccator, all specimens shall be brought to room temperature in a second desiccator in which a similar vacuum is maintained. After reaching room temperature, they shall be removed and weighed immediately with an accuracy of one part in 10,000. These specimens from the insulated conductors shall be bent in the shape of an "U" around a mandrel not less than three times the diameter of the insulation. The specimens shall be supported by fitting the ends snugly through holes in a metal plate which shall serve as the cover for a vessel of distilled water. The specimens shall be of such length and so adjusted that not less than a nine-inch length of the specimen is exposed below the plate and not more than one-half inch length of each end is exposed above the plate. Care shall be taken in this end subsequent procedure that no liquid gains access to the stranded conductors. The insulated conductors shall fit tightly in the cover and the cover shall fit tightly to the vessel to prevent excessive evaporation. The level of the water shall be maintained flush with the under side of the cover, and steps shall be taken to eliminate the possibility of air gaining access to the water or inside of the vessel. All surface below the under side of the cover shall be considered as immersed. The area of the surface immersed shall be taken as the product of the length immersed and the average circumference as measured at the middle and two inches from each end of each specimen. All measurements shall be taken prior to immersion, with an accuracy of one part in 100.

4.4.2.2 Immersion period.- The vessel of distilled water containing the specimen shall be maintained at a temperature of not less than 69°C. or more than 71°C. Specimens shall remain immersed for a period of 168 hours (seven days). At the end of this period, they shall be prepared for weighing by bringing them to room temperature in distilled water. As each specimen is about to be weighed, it shall be removed from the water and the loose water shaken off. It shall be immersed immediately in alcohol for two seconds, removed, shaken and blotted with lintless absorbent cloth. The alcohol used shall be pure 95 percent ethyl alcohol, or the same denatured with pure methyl alcohol. (The purpose of the blotting is to remove the surplus alcohol or

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water adhering to the surface, and should be accomplished by placing the specimen between loose folds of cloth, employing as little friction as possible.) The blotting process should be completed within 30 seconds after removal from the alcohol. The specimen shall be placed on the balance and the weight recorded 3.0 minutes after removal from the alcohol. After weighing, the specimens shall be returned to the desiccator and maintained under the conditions specified in 4.4.1. They shall be removed from the desiccator and reweighed at intervals of 24 hours until constant or minimum weight is obtained. In each case they shall be prepared for reweighing by first bringing them to room temperature in another desiccator in which a similar vacuum is maintained. The gain in weight during the 168 hour immersion period and the loss in weight during the redrying period shall be recorded. If the loss in weight exceeds the gain in weight by more than 20 percent, it shall be deemed evidence of excessive leaching and the material represented by the sample shall be rejected. The decrease in weight during the period of redrying shall be taken as the weight of water absorbed during immersion. The water absorbed shall be reported in milligrams per square inch of surface immersed.

4.4.3 Electrical water absorption test.-

4.4.3.1 Test specimens.- Test specimens shall be made up into a coil with an inside diameter of not less than two inches. The specimens shall be supported by fitting the ends snugly through holes in the cover of the vessel of tap water. The specimens shall be of such length and so supported that not less than five feet of conductor is exposed below the cover, and that the ends do not project more than six inches above the cover. The insulation shall be removed from the ends for a distance of one-half inch. The water shall be maintained at a temperature of not less than 69°C. nor more than 71°C. The conductors shall fit tightly in the cover, and the cover shall fit tightly to the vessel, to prevent excessive evaporation. The level of the water shall be maintained flush with the underside of the cover, and steps shall be taken to eliminate any possibility of air gaining access to the water or inside of the vessel. The length of the conductor below the under side of the cover shall be taken as the length immersed.

4.4.3.2 Measurements.- If the vessel is made of metal, it shall be considered the electrical ground wire. If it is not of metal, a ground plate of suitable area shall be immersed in the water. All metal parts other than the conductor under test shall be grounded. The capacity shall be measured between the conductor and the ground, at 1000 cycles per second, with a suitable alternating current bridge, with an accuracy of one micro-microfarad of capacity and 0.001 power factor. All measurements shall be made at 70°C. without distrubing the test specimen in any manner. The capacity shall be recorded in microfarads per 1000 feet of conductors immersed. The first measurement shall be made after 24 hours of immersion at 70°C. and shall be considered the original capacity. Subsequent measurements shall be made at

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intervals of seven and 28 days after the original measurement. The power factor shall be measured at the end of the first seven day period.

4.5 Tests on all entire lengths.-

4.5.1 Conductors.- Each completed length of insulated conductor after vulcanization and before the application of any covering, shall be reeled through water and immediately immersed in fresh water at room temperature for a period of not less than 16 hours, after which period of time and while still immersed it shall be subjected to the voltage, insulation and capacity tests as specified in 4.5.4, 4.5.5 and 4.5.6.

4.5.2 Intermediate (core) test.- Each completed length shall be immersed in fresh water for a period of not less than eight (8) hours and shall be subjected to the voltage and insulation test specified in 4.5.4 and 4.5.5. Each individual length that meets the specified requirements may be accepted. Each length that fails shall be rejected, but may be repaired in accordance with 4.5.7 and retested. For the purposes of this paragraph a length of core shall consist of all insulated and cabled conductors together with tapes, fillers, jackets, and other protective coverings as may be specified, which are applied prior to the armoring operation. If a lead sheath is to be permanently applied or an outer jacket is applied, the core may be tested dry.

4.5.3 Completed cable.- Each completed length of cable shall be immersed in fresh water for a period of not less than 48 hours and shall be subjected to the voltage and insulation test specified in 4.5.5 and 4.5.6. In making all electrical tests, all metal parts of the cable and all conductors not under test shall be grounded. Each individual length that meets the specified requirements shall be accepted. Each length that fails shall be rejected, but may be repaired in accordance with 4.5.7 and retested.

4.5.4 Carload lengths.- If the contract calls for delivery of the cable either spliced or armored in carload lengths and coiled in a gondola car, each carload length shall be subjected to the voltage and insulation resistance tests specified in 4.5.5 and 4.5.6, before shipment. Each length may be wet or dry.

4.5.5 Voltage tests.- AC or DC potentials as specified in 3.3.2 shall be applied between the conductor or conductors under test and ground for the periods specified. At the discretion of the inspector, alternate conductors in the same layer may be divided into two groups, and a test made between each group and ground, with the group not under test grounded. If an AC potential is specified, the frequency shall be not less than 25 nor more than 100 cycles per second.

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4.5.6 Insulation resistance test.- The test shall be made with a direct potential of not less than 200 volts, applied for a period of one minute between the conductor under test and ground. A galvanometer of suitable sensitivity shall be inserted in the circuit to indicate the current. The resistance shall be computed by comparing the deflection of the galvanometer after the potential has been applied to the cable for one minute with the deflection obtained when the same potential is applied to a standard of known resistance. The result shall be expressed in megohms per 1000 feet, and, when corrected to a temperature of 60°F., shall be within the limits specified. If required by the inspector, tests shall be made to determine the temperature coefficient to be applied in making corrections. In any case the contractor shall furnish a table of temperature coefficients covering the range from 40°F. to 90°F. for the compound furnished.

4.5.7 Repairs.- Lenghts of completed cable that fail to meet the tests specified in 4.4.2 due to a localized fault or faults may be repaired and reoffered for inspection. Where repairs are made to the cable, the work shall be performed in such a manner that all parts affected in the process shall be as strong and durable electrically and mechanically as the remainder of the cable. The armor may either be spliced in accordance with Section 5 of this specification or may be laid back in its original position and electrically butt welded or butt brazed. If the armor is welded, the joints in the several armor strands shall be staggered over a distance of not less than 30 feet, and the welds shall be thoroughly coated with zinc. When class 1 or 2 jackets are specified, the number of repairs or splices shall not exceed one for each length of completed cable not exceeding 5000 feet in length, with one additional repair or splice permitted for each additional 5000 feet or fraction thereof, provided that no two armor splices shall be less than 1000 feet apart. When class 3 or 4 jackets are specified, splices may be made as close together as 2000 feet.

5. PREPARATION FOR DELIVERY

5.1 Factory splicing.- When class 1 covering or class 2 jacket is specified, a completely manufactured cable shall contain no factory splice in the first 5000 feet of cable and no more than one factory splice in a total continuous length up to 7500 feet. Longer cables may have factory splices spaced no closer together than 5000 feet with none permitted in the initial 5000 feet. When cable is supplied with class 3 or 4 jackets, splices are permitted as close together as 2000 feet. Repair splices, as distinguished from factory splices, shall be in accordance with 4.5.7.

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5.1.1 Splicing diagram.- If the order provides for cable to be spliced at the factory before shipment, either before or after armoring, the order in which the lengths are to be spliced together and the arrangement of the conductors in the splices shall be determined from a splicing diagram, which shall be furnished to the contractor by the purchaser or designated inspector upon completion of the required balancing. All splices shall be made in accordance with this splicing diagram.

5.1.2 Methods of factory splicing.- On all cable splices made by the factory, the conductors (and metal tapes when specified) shall be butt brazed or soldered. The insulation and jackets shall be vulcanized in a suitable manner. The splices shall be so made that all parts affected in the process shall be as strong and durable, electrically and mechanically as the remainder of the cable.

5.1.3 Armor splicing.- On cable spliced after armoring, the armor splice may be made by laying back the wires in their original position and electrically butt welding as specified in 5.1.2 to form a complete splice, or the armor wires may be laid back in position one at a time parallel with the splice. As an armor wire from one side is brought into position, one from the opposite side should be laid in place beside it, so that they will interlock in the center of the splice. Care should be taken to remove the sharp bend at the temporary serving. The wire should be evenly distributed around the splice and squeezed into position if necessary with cable splicing clamps to form a smooth round splice, as small in diameter as possible. Place a serving of 10 turns of 0.134-inch diameter galvanized iron serving wire around the center of the splice, drawing it as tightly as possible with the serving stick. On each side of this serve, select four armor wires (six if the armor is light) spaced equidistantly around the splice, bend them back over the center serving, and hammer and squeeze them down snugly with the other armor wires to form an interlocking joint. Serve the balance of the splice with 0.134-inch diameter galvanized iron serving wire, drawn as tightly as possible with the serving stick. If the armor is smaller than 0.134-inch diameter, the serving wire shall be of the same diameter as the armor. To start the serving, loop the serving wire around one of the armor wires and lay the end parallel with the armor so that it will project beyond the farther end of the serving. When the serve is completed, the two ends of the serving wire should be twisted together for an inch and hammered down parallel with the serve. Start each serving as close as possible to the center serving, and serve towards the ends of the splice. On jacketed cable, the final turn should be made and secured about one inch before the end of the jacket is reached, taking care that the armor wires do not injure the jacket while serving. On non-jacketed cables, the serve should overlap the original jute an inch. Bend the armor wires back over the serving toward the lockjoint, hammer them flat against the serving,

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and cut them off as short as possible.

5.1.3.1 Serving stick.- The serving stick or device used shall be so constructed that it does not materially injure the galvanized coating of the serving or armor wire during the serving operation.

5.1.3.2 Numbering.- A consecutive number will be assigned for each splice on the splicing diagram. Each splice shall be marked in duplicate with this number by means of bands of iron wire placed on the cable at points three feet from each end of each splice. The number of turns of iron wire so placed shall coincide with the number of the splice. In order to facilitate counting, the turns shall be applied in groups of five turns with a one inch spacing between groups. Example: Splice No. 12 would consist of two bands of five turns each and one band of two turns.

5.1.4 Testing vulcanized conductor joints.- After completion of jointing and vulcanizing the conductors, and before replacing the jute and armor, they shall be immersed in water and a DC voltage applied for five minutes in accordance with the DC voltage limits specified in 3.3.2.

5.2 Reel shipment.-

5.2.1 General.- When the order provides for cable to be placed on reels, the actual lengths of the cable on reels, after the final inspection at the factory, shall not be less than specified in the order. Unless otherwise specified, no more than 5000 feet of Type I cable shall be furnished on an individual reel (plus 5 percent tolerance in length). Each length shall be wound on a separate reel unless otherwise stated. The reels shall be substantial and able to withstand such reasonable handling as they are liable to receive in transit. The diameter of the drum shall be large enough to prevent damage to the cable. The head of the reel shall be provided with a slot, through which the inner end of the cable will be accessible for testing. The outer end of the cable shall be securely fastened to the inner side of the cable head so that the cable will not come loose in transit. All reels shall be enclosed with wooden lagging securely attached to the outside flanges of the reel.

5.2.2 Marking.- The reel shall be plainly marked to indicate the direction which it should be rolled so as not to loosen the cable on the reel. The head of each reel shall be stenciled or lettered with waterproof ink or paint as follows:

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Contractor's name and address

Name and address of consignee

Contract and item number

Length and type number of cable

Reel number

Net weight of cable

Net weight of reel

Expiration date reel returnable for credit
(Month-Year) (as applicable)

5.3 Carload shipment.- When the order provides for shipment of cable in one length or various lengths in carload lots (gondola cars), the actual length or lengths of the cable after final inspection at the factory shall not be less than specified in the order. The cable shall be coiled in flat even layers, starting at the outside and working toward the center of the coil, stopping at a point where turns will not lie flat and excessive bending will become injurious to the cable, the diameter of the inner coil being dependent on the size and type of cable being coiled. The cable should then cross over to the outside again and successive layers be made in a similar manner. The cross-overs shall be distributed so that they do not all appear on one side of the coil. Pieces of dunnage slightly thicker than the diameter of the cable should be placed alongside the cable on each cross-over so that excessive weight will not fall on this part of the cable. The ends of the cable shall be brought out clear of the coil for testing purposes. The cable shall be properly braced with shoring and anti-abrasion protection to prevent shifting. The cable shall be properly roofed in order to minimize fire hazards and other injuries to the cable. One end of each length of cable shall have a tag securely attached showing the following information:

Contractor's name and address

Name and address of consignee

Contract and item number

Quantity and type number of the cable

Weight of cable

Length of cable

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6. NOTES

6.1 Intended use.- The cables covered by this specification are intended for submarine applications in conveying power and controlling off-shore aids to navigation and other installations.

6.2 Ordering data.- Procurement documents should stipulate the following:

(a) Title, number and date of this specification.

(b) Length of cable required.

(c) Type required (see 1.2).

(d) Whether conductors shall be shielded (see 3.2.1.2).

(e) Whether armor wires shall be tar-dipped (see 3.1.2.4).

(f) Whether bronze tape shall be included (see 3.1.8, 3.2.4.1, 3.2.4.2, 3.2.4.3, 3.2.4.4, and 3.2.4.5).

(g) Size of armor (see 3.2.4.1(c) and 3.2.4.2).

(h) Whether lead sheath shall be included (see 3.2.4.5.6).

(i) Polyethylene jacket compound (see 3.1.6.1).

(j) Length of cable per reel and whether wood reels or steel reels are desired. If wood reels are desired, whether returnable or non-returnable.

6.3 Bid requirements.- Each bid or quotation shall be accompanied by a complete detailed description of the proposed cable, showing design and insulating compound. Failure to furnish the above information shall be non-responsive. Bidders shall not refer to previous Government contracts for this descriptive information.

6.4 General .- When the contract is awarded, the Government will inform the contractor of the name and address of the officer designated to make the inspection, the contractor shall notify the officer so designated by mail or telegraph of the date on which inspection is desired. Such notice shall be forwarded to reach the inspector not less than five government working days prior to the day on which the inspection is to commence. The inspector shall

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acknowledge receipt of the notice and inform the contractor of the date of his intended arrival. If five days advance notice has been given and inspection is postponed at request of the inspector, delivery time designated in the contract shall be extended by the number of days the inspection is postponed. If any material or cable is rejected by the inspector, his decision shall be deemed final unless appealed by the contractor to the purchasing officer within 48 hours.

NOTICE.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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INDEX

Aging 4.3.2
Armor
Material 3.1.2
Size 3.1.2.1
Zinc coating ..... 4.3.2 thru 4.3.2.2
Tensile strength and test of 3.1.2.2
Application in cable 3.2.4.6.2
Bronze tape (for protective covering) 3.1.8 and (c) of 3.2.4.2 thru 3.2.4.5
Bid requirements 6.3
Cabling
General 3.2.4
Coverings ..... 3.2.4.1 thru 3.2.4.5
Details
Inner jute serves 3.2.4.6.1
Armor 3.2.4.6.2
Outer jute serves (for class 1 coverings only) 3.2.4.6.3
Synthetic rubber jacket (class 2, 3, and 4 coverings) 3.2.4.6.4
Seine twine (classes 3 and 4 coverings only) 3.2.4.6.5
Lead sheath 3.2.4.6.6
Capacity 3.3.3
Carload shipment 5.3
Characteristics, Electrical
Capacity 3.3.3
Insulation resistance 3.3.1
Voltage requirements 3.3.2
Classifications
Cables 1.2.1
Protective coverings .....1.2.2, 3.1.6, 3.1.7, 3.2.4.1 and 3.2.4.6.4
Tests 4.2
Color coding 3.2.3
Compounds
Conductor insulating 3.1.5
Protective jacket 3.1.6
Tar dip (for armor wire) 3.1.2.4
Tar and asphalt (for outer covering of Type I cables) 3.1.7
Compression cutting test 4.3.3

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Conductors
Material 3.1.1
Size and ratings 3.2.1
Stranding 3.2.1.2
Shielding 3.2.1.1
Insulating 3.2.2
Color coding 3.2.3
Testing
On samples ..... Table V
On all complete lengths ..... 3.3.2 (1) & 4.4.1
Construction ..... 3.2 thru 3.2.4.6.6
Copper corrosion test 4.3.4
Delivery, preparation for
Splicing ..... 5.1 thru 5.1.4
Marking 5.2.2
Reel shipment 5.2.1
Carload shipment 5.3
Documents, applicable ..... 2.1 thru 2.2
Elongation ..... Table III & Table V
Inner jute serves 3.2.4.6.1
Insulation compound 3.2.2
Insulation resistance ..... 3.3.1 & 4.5.6
Jacket ..... 3.1.6.1 & 3.2.4.6.4
Jute, general ..... 3.1.4 to Table II
Jute, vacuum impregnated
Material 3.1.4.2
Layers in cabling ..... 3.2.4.1 thru 3.2.4.3
Jute, waterproof
Material 3.1.4.1
Layers in cabling ..... 3.2.4.1 thru 3.2.4.5
Lead sheath 3.2.4.6.6
Marking 5.2.2
Materials ..... 3.1 thru 3.1.11
Outer jute serves (for class I, covering only) 3.2.4.6.3
Ozone resistance ..... 3.2.2 & Table V
Reel shipment 5.2.1
Repairs, cable 4.5.7
Sampling for tests ..... 4.3 and Table V
Seine twine (classes 3 and 4 coverings only) 3.2.4.6.5
Shielding tape 3.2.1.2
Splicing ..... 5.1 thru 5.1.4
Standards referenced
Federal 2.1
Commercial 2.2
Test specimens 4.3.3.1, 4.3.4.1, 4.4,1, 4.4.3.1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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